SUSPENSION—VEHICLE APPEARS TO DOGTRACK—SERVICE TIP	Article No. 02-16-8

FORD: 1992-2003 E SERIES

Article **00-5-10** is being republished in its entirety to update the vehicle model years.

ISSUE

The customer may report that an E-Series appears to dogtrack. This customer concern is generally due to unique features of an E-Series that combine to create an optical illusion of dogtracking. Dogtracking is the condition in which the rear axle is not square to the chassis. Heavily crowned roads can add to the illusion of dogtracking.

ACTION

Become aware of the unique features that can create the optical illusion of a dogtrack condition. Follow the Service Procedure to determine if a dogtrack condition exists. If a dogtrack condition can be verified, follow the Diagnostic Procedure and Repair Instructions to correct the condition.

SERVICE PROCEDURE

NOTE
DOGTRACK CONDITIONS THAT ARE THE RESULT OF A COLLISION ARE NOT TO BE ADDRESSED USING THIS TSB. DOGTRACKING THAT IS THE RESULT OF A COLLISION MUST BE REPAIRED USING GENERALLY ACCEPTED COLLISION REPAIR TECHNIQUES.

NOTE
NEARLY ALL CUSTOMER REPORTS OF DOGTRACKING ARE THE RESULT OF THE OPTICAL ILLUSION DOGTRACKING THAT IS COMMON ON THE E-Series.

Two unique conditions combine on the E-Series that enhance the optical illusion of dogtracking.

First: Front wheel track center of tire to center of tire is wider than the rear wheel track. A person traveling behind the vehicle will be able to see both the front wheel and the rear wheel. The rear wheel will appear to be further toward the center of the vehicle (Figure 1). Front wheel track = 1.76m (69.4″), rear wheel track = 1.70m (67.0″).

Second: The body of the vehicle is widest in the middle and tapers in toward the front and rear of the vehicle. The rear width of the vehicle is nearly 10cm (4 ″) narrower than the widest section of the vehicle in the middle. The SuperVan (extended length) continues to taper to the rear of the vehicle and magnifies the optical illusion of dogtracking. Because of the taper of the body, a person driving behind the vehicle can see the side of the vehicle. ONLY if a true condition of dogtracking is suspected, follow Service Procedure to verify and repair.

NOTE
ATTEMPTING TO USE THE FOLLOWING SERVICE PROCEDURE TO REDUCE THE APPEARANCE OF DOGTRACKING ON A VEHICLE THAT IS NOT TRULY DOGTRACKING WILL LEAD TO ALIGNMENT AND TIRE ISSUES. DO NOT USE THIS PROCEDURE UNLESS A TRUE DOGTRACK CONDITION IS VERIFIED!

NOTE
SOME NON-FORD AFTERMARKET SPRINGS DO NOT HAVE A CENTER BOLT IN THE LEAF SPRING, WHICH CAN CAUSE DOGTRACKING BY NOT CENTERING THE AXLE ON THE LEAF SPRING. UNEVEN LOADING OF THE VEHICLE (LEFT TO RIGHT) CAN ALSO CAUSE DOGTRACKING.

DIAGNOSTIC PROCEDURE

1. Verify the vehicle dogtracks by measuring the rear axle thrust angle (Figure 2) on the vehicle with a calibrated 4-wheel alignment machine.
2. Record reading and follow repair instruction based on reading:
   a. Thrust angle of 0 to ±0.5 degrees
      (1) Thrust angle less than 0.5 degrees (32mm/1.25″centerline difference) is in specification
   b. Thrust angle of ±0.51 to 0.8 degrees
      (1) To modify thrust angle by 0.1 to 0.3 degrees:

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(a) Modify the spring center bolt head as shown in Figure 3, on the side of the axle which must be moved. If the vehicle dogtracks to the right, grind the rear of the right side leaf spring pin. Reverse for dogtracking to the left. Follow repair instructions.

(b) Adjust the axle by shifting it forward to make contact with the ground pin prior to torque down.

c. Thrust angle of ±0.81 to 1.0 degrees

(1) To modify thrust angle by 0.3 to 0.5 degrees:

(a) Modify the spring center bolt head as shown in Figure 3, on both sides of the axle. If the vehicle dogtracks to the right, grind the rear of the right side leaf spring pin and the front of the left side leaf spring pin. Reverse for dogtracking to the left. Follow repair instructions.

(b) Adjust the axle so that both ends make contact with the ground pins prior to torque down.

d. Thrust angle of ±1.01 or greater

(1) To modify thrust angle by more than 0.5 degrees:

(a) Ensure that 4-wheel alignment is correctly calibrated and re-measure.

(b) If reading is the same, have the frame measured and follow generally accepted collision repair procedures to straighten frame.

REPAIR INSTRUCTIONS

NOTE
FOLLOW NORMAL AND CUSTOMARY SAFETY PRECAUTIONS WHEN LIFTING A VEHICLE. TO REVIEW THE SAFETY PRECAUTIONS, REFER TO SECTION 100-02 OF THE E-Series 1998 TO 2003 WORKSHOP MANUAL.

1. Raise the vehicle off the ground using a frame-mounted hoist or equivalent. Do not hoist the vehicle using the rear axle as a lift point, as you will need the axle to be free for the repair.

NOTE
PERFORM REPAIR ON ONE SIDE AT A TIME IF ADJUSTMENT TO BOTH SIDES IS NECESSARY.

2. Support the rear axle with jack stands and then remove the U-bolts which attach the rear axle to the rear leaf springs on the side of the axle which will be moved. If the thrust angle is large, both sides of the axle may require adjustment (see diagnostic procedures). Remove the lower shock absorber and bolt from the axle and lower the jack stand on the side of the axle to be adjusted.

3. Using C-clamps, clamp the leaves of the leaf spring together on both sides of the center bolt and remove the bolt (Figure 4).

4. Reinstall the center bolt and torque the nut to 70 ±10N•m (52 ±7 Lb-ft). Push the axle the maximum distance in the direction required to reduce the thrust angle and tighten the U-bolt nuts: 115.5 ±17.5N•m (85 ±13 Lb-ft) for 14mm nuts, 175 ±27 N•m(129 ±20 Lb-ft) for 16mm nuts.

5. Reattach the shock absorber and torque to 79N•m (58 Lb-ft).

6. Verify the vehicle no longer dogtracks by rerunning the Diagnostic Procedure.

SUPERSEDES: 00-5-10
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
021608A	Correct Thrust Angle Or Rear Axle	1.6 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
4210	42

OASIS CODES: 303000, 304000, 305000, 390000

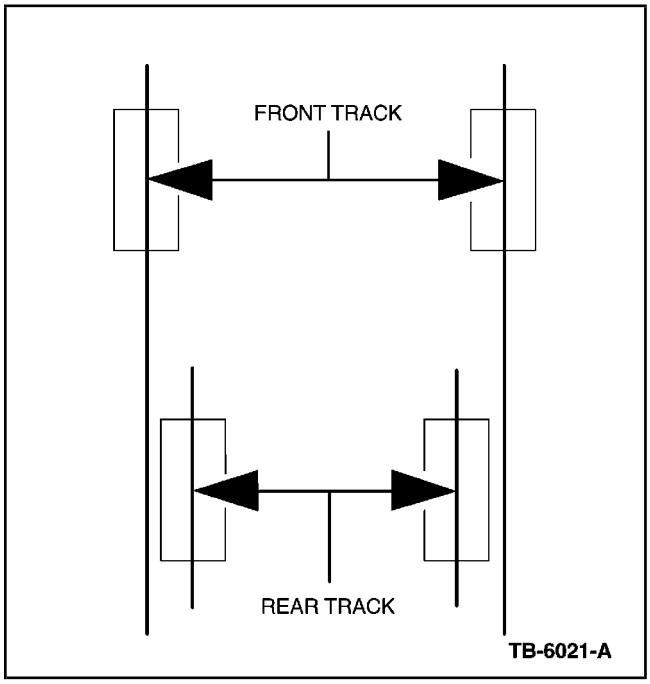


Figure 1 - Article 02-16-8

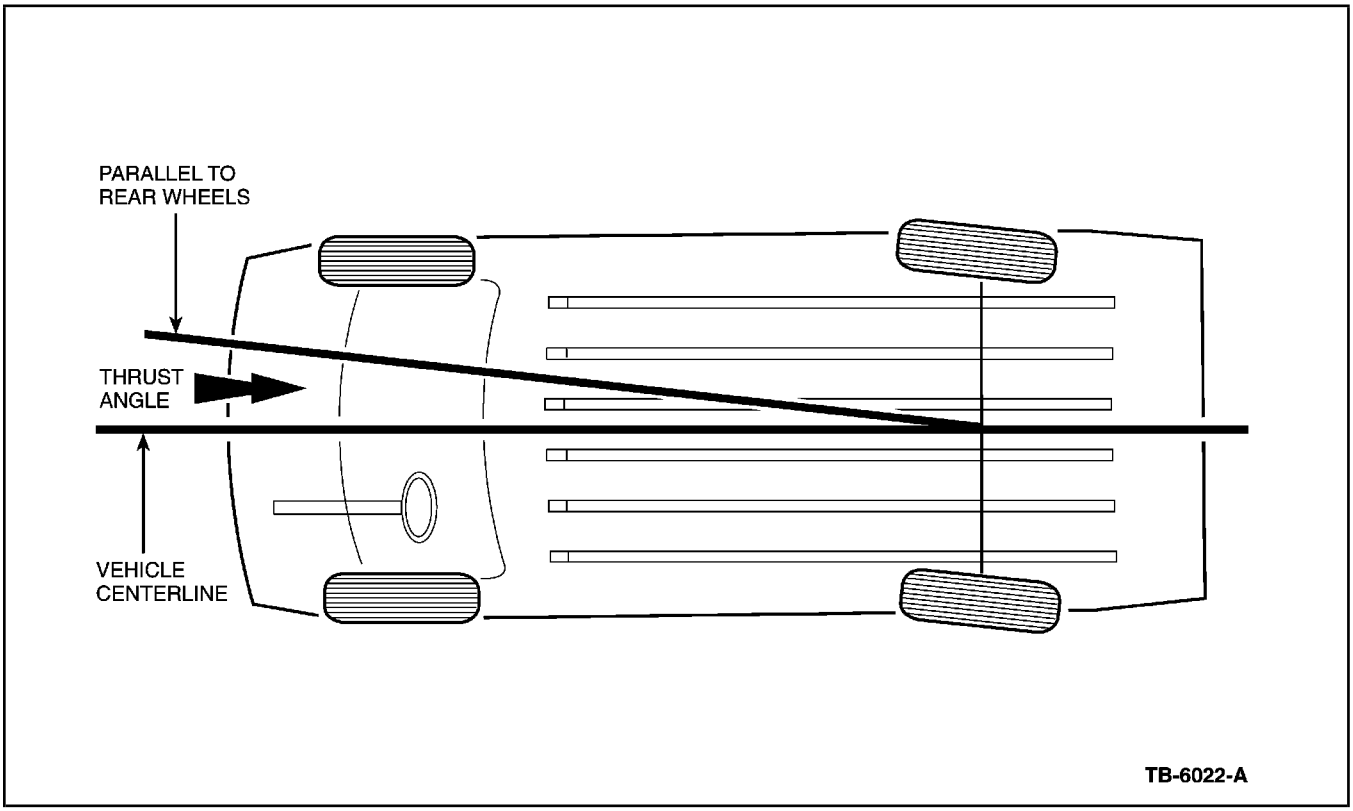


Figure 2 - Article 02-16-8

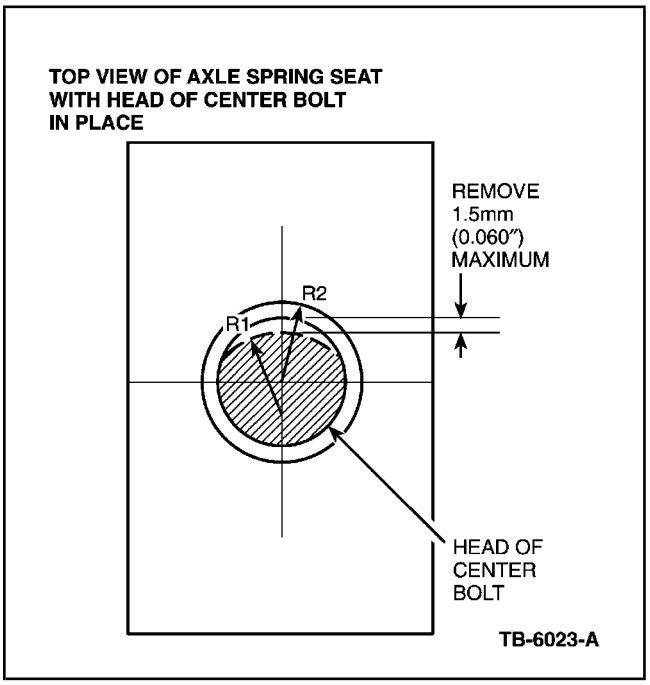


Figure 3 - Article 02-16-8

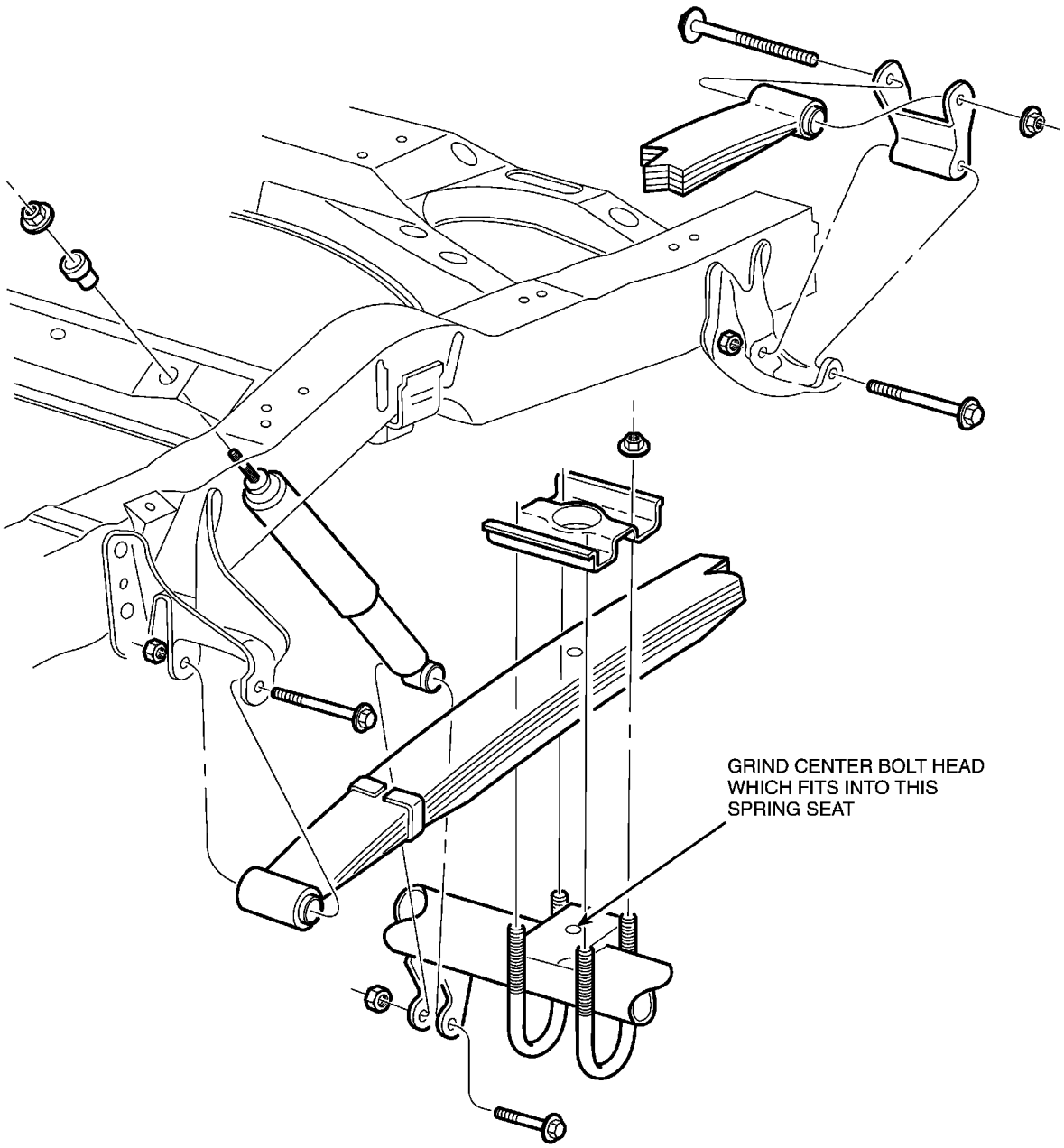


Figure 4 - Article 02-16-8